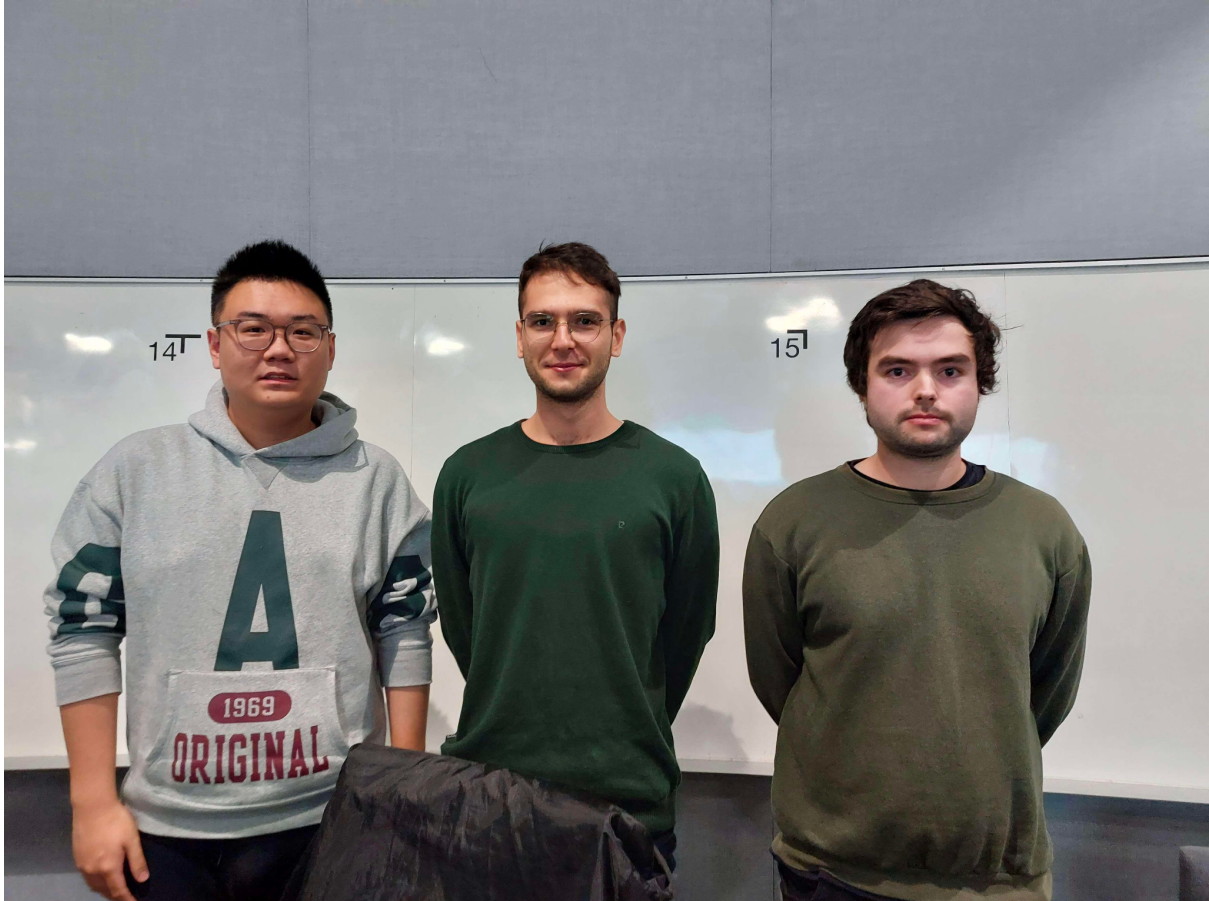


SPRINT ONE DELIVERABLES

Team Name: ForThe3

Team Photo:



Team Membership:

Thomas Hogarth:

Email - thog0001@student.monash.edu

Strengths: Pattern recognition and problem solving

Fun fact: My grandmother worked at Monash when I was a child.

Han Zhu:

Email - hazu0048@student.monash.edu

Strengths: I have analytical thinking and enthusiasm for working together.

Fun fact: The best advice I ever received was whatever you do—give it 100%

Ozan Kaymak:

Email - okay0002@student.monash.edu

Strengths: Strong mathematical background

Fun fact: I am an Exchange student from Turkey, and I changed my address here to be able to vote.

Team Schedule:

Meetings: Two meetings per week, one on Monday morning (10:00 a.m.) and the other one on Friday morning (10:00 a.m.).

Task Management: Distribution of work will be discussed during meetings. All members will be contributing in every aspect of the project, not necessarily heavily, but at least with familiarity and if needed, help others to solve problems. Task allocation will be recorded in a google doc, and updated as the project progresses.

Technology Stack and Justification: The project will be built in Java, using the Swing api to create the graphical user interface. All team members are comfortable building with Java, and prefer the simplicity of the Swing toolkit, rather than the more complex, detail oriented Java GUI frameworks.

The other consideration was to build in Python, using wxPython or similar GUI toolkit api. However Java's constraint as a purely object oriented language, will ultimately aid in the development of the project.

Lucid link:

https://lucid.app/lucidchart/db999a4c-5c84-40c2-b832-0b7ff9bbf6ae/edit?viewport_loc=-5%2C-25%2C2228%2C1272%2C0_0&invitationId=inv_8bbae1f5-7b8d-4c60-b197-509d2ba2f3f9

Figma link:

<https://www.figma.com/file/9xn1bxgPiOoqMhX05eFnly/Ozan-Kaymak's-team-library?t=b7svumFFRLFvrGIS-0>

User Stories:

As a/an game player,
I want to know/learn the game rules,
So that I can develop a strategy for the game and win.

As a/an game player,
I want to see my remaining pieces.
So that I can know if I should put new pieces or move pieces.

As a/an game player,
I want to see what stage of the game myself and my opponent are currently in,
So that I can know what plays we're allowed to make and strategize.

As a/an game player,
I want to see the game state clearly,
So that I can understand the position the game currently is in.

As a/an game player,
I want to see the remaining pieces of my opponent,
So that I can understand what kind of strategies are possible for my opponent.

As a/an game player,
I want to see a visual indication of my opponents move,
So that I will not be confused as to how the board changed state.

As a/an game player,
I want to have a simple but elegant design for the game interface,
So that my eyes will not be tired and I will not get bored.

As a/an game player,
I want the game to have a visual flow for the moves,
So that the game is eye-catching for me.

As a/an game player,
I want the game to not have any delays after moves,
So that I can comfortably play the game.

As a/an game board,
I want to the game to have a very clear status display,
So that I can show the game steps.

As a/an game viewer,
I want to see how many pieces each player has captured,
So that I can more accurately gauge who is winning.

As a/an game viewer,
I want to see how many pieces each player has unplaced,

So that I can see which state the game is in.

As a/an new player,

I want to have the option to see all available moves for a selected piece,

So that I do not get lost and ruin the playing experience.

As a/an token,

I want to have a clear symbol,

So that I can be seen by the players.

As a/an position,

I want to be able to be marked as occupied by a token,

So that no tokens will overlap.

As a/an game board,

I want to confirm that each step piece moves as specified,

So that a fair game can be played.

As a/an advanced player,

I want to be able to skip the tutorial (if there is one),

So that I don't waste my time and start playing the game directly.

As a/an hint,

I want to have access to all possible moves for a token,

So that I know which moves were valid/invalid.

As a/an hint,

I want to provide help to player that toggled me,

So that I can assist in the game play.

Design Rationale

Game will act as the engine which provides players, turns, and will call their moves using the user's input. Keeping Game top level and readable, makes the entire system easy to understand and extend. The design of this project, places heavy emphasis on the Single Responsibility Principle. For example:

Player is only responsible for using tokens to make moves.

Token is only responsible for knowing their owner (plus owner's game stage) and position.

Move is only responsible for validating player moves.

Board is only responsible for updating the state of tokens and their positions.

Hint is only responsible for finding all valid moves for a player's token.

Position is only responsible for knowing its current state.

The emphasis on SRP requires more associations between entities to maintain this. This has its drawbacks, specifically increasing visibility, but the level of readability and ease of extension gained from SRP, justifies the style of design.

Game: The project requires a Game entity, to create two Players and execute moves based on the user's input. This game must also have one Board, whose information it can pass to the Display.

Board: A board must exist to keep track of every new Move, all of the Tokens and their Positions. This information is crucial for the Game to receive and Display.

Player: A player is needed to, declare ownership of, *and use* their Tokens to make Moves.

Token: A Token is required to belong to one of two Players and a single Position on the board (if it has been placed already).

Move: Move is needed to facilitate the transfer of one Token from its current Position to another valid Position on the board. The player can directly make a Move of Type related to their current game stage (explained in more detail below).

Position: A Position must exist to identify as a single place on the board, which borders other Positions on the board.

Hint: The project requires a Hint entity so that it can find all valid potential Moves for a player's Token.

Display: Display is needed to take board state information from the Game, so that the user can see each of the three types of display.

TutorialDisplay: TutorialDisplay uses unique visual elements to teach the Player the rules of the game, and how to navigate the UI.

MessageDisplay: MessageDisplay prints the latest change to the board state as a message to the screen.

BoardDisplay: BoardDisplay displays the current state of the board to the screen.

MoveType: MoveType organizes the three unique move styles belonging to the respective game stage the player is currently in. The Player can use the appropriate MoveType for their game stage.

Put, Slide, Jump: Each of these entities provide the tools to validate a player's move of *that* respective type.

Assumptions/Further Justification:

Modeling the board as an organized collection of positions, instead of a collection of lines made up of positions, makes the check for formed mills more complicated. We are still considering whether this warrants the inclusion of a new Line entity.

The removal of a token after forming a mill is a two step process, which is realized within the Move entity and its associations. We chose not to model this in this design because it is more appropriate to be fleshed out in a class diagram.